

Integration

- the **opposite** of differentiation
- can be used to find the **area under a curve**

1. The Basics

- a) The power rule: (i) **increase power by 1** — then divide by it
 (ii) stick a **constant** on the end

eg: $\int x^3 dx = \frac{1}{4} x^{3+1} + c = \frac{1}{4} x^4 + c = \frac{x^4}{4} + c$

$\int \frac{1}{x^3} dx = \int x^{-3} dx = \frac{1}{-2} x^{-3+1} + c = \frac{1}{-2} x^{-2} + c = \frac{1}{-2x^2} + c$

'indefinite integral'
 no limits (numbers)
 next to integral sign

$$\int x^n dx = \frac{1}{n+1} \cdot x^{n+1} + c$$

a constant

$n \neq -1$

$\int 4x^2 dx = 4 \int x^2 dx = 4 \times \frac{1}{3} x^{2+1} + c = 4 \times \frac{x^3}{3} = \frac{4x^3}{3}$

a constant

$$\int k f(x) dx = k \int f(x) dx$$

b) constant multiples

c) sums and differences

$$\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$$

$\int 5x^3 + 4 dx = \int 5x^3 dx + \int 4 dx = \frac{5x^4}{4} + \frac{4x^1}{1} + c$

2. Finding the constant c , given: - the gradient function
- a coordinate on the curve

e.g.: curve y , where $\frac{dy}{dx} = \frac{6x^5 - 18}{x^3}$, point $(1, 6)$

$$\frac{dy}{dx} = 6x^2 - 18x^{-3} \quad \leftarrow \text{index form ready for integration}$$

$$y = 2x^3 + 9x^{-2} + c$$

$$= 2x^3 + \frac{9}{x^2} + c$$

substitute $x = 1$, $y = 6$

$$6 = 2(1)^3 + \frac{9}{(1)^2} + c$$

$$6 = 2 + 9 + c$$

$$c = -5$$

$$\longrightarrow y = 2x^3 + \frac{9}{x^2} - 5$$

3. For $(ax + b)^n$ expressions

only works when inside $()$ is linear a.k.a.: x^1

$$\int (ax + b)^n dx = \frac{1}{a(n+1)} (ax + b)^{n+1} + c$$

$a \neq 0$

$n \neq -1$

$$\text{e.g.: } \int (3x - 1)^6 dx = \frac{1}{7 \times 3} (3x - 1)^7 + c$$

4. Further indefinite integration

a Show that $\frac{d}{dx} [(3x^2 - 4)^8] = 48x(3x^2 - 4)^7$.

b Hence, find $\int 6x(3x^2 - 4)^7 dx$.

Answer

a Let $y = (3x^2 - 4)^8$

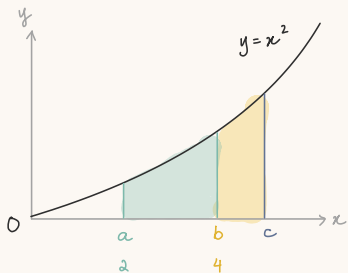
$$\begin{aligned} \frac{dy}{dx} &= (6x)(8)(3x^2 - 4)^{8-1} \\ &= 48x(3x^2 - 4)^7 \end{aligned}$$

Use the chain rule.

b
$$\begin{aligned} \int 6x(3x^2 - 4)^7 dx &= \frac{1}{8} \int 48x(3x^2 - 4)^7 dx \\ &= \frac{1}{8} (3x^2 - 4)^8 + c \end{aligned}$$

$$\text{If } \frac{d}{dx} [F(x)] = f(x), \text{ then } \int f(x) dx = F(x) + c$$

5. Definite integration



e.g. $\int_2^4 x^2 dx = \left[\frac{1}{3} x^3 + c \right]_2^4$ has limits

$$= \left(\frac{1}{3} \times 4^3 + \cancel{c} \right) - \left(\frac{1}{3} \times 2^3 + \cancel{c} \right)$$

$$= \frac{56}{3}$$

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$$

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

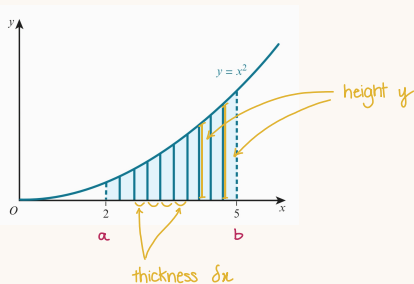
$$\int_a^b k f(x) dx = k \int_a^b f(x) dx$$

all stuff you've seen before,
but with limits

$$\int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$$

6. Area; under a curve

- divide area into slices of equal thickness:



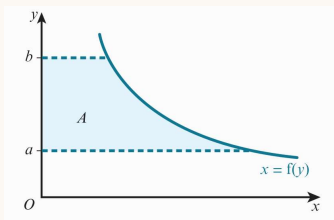
The approximation of A is $\sum y \times \Delta x$ area

As $\Delta x \rightarrow 0$, then $A \rightarrow \int_2^5 y dx$

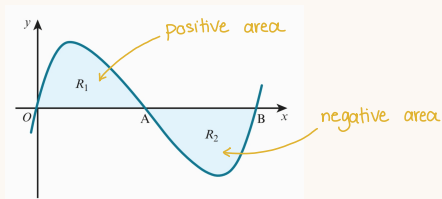
the total area is divided into equally thinner slices,
so thin! thickness is practically 0

$$A = \int_a^b y dx$$

• area between curve and x-axis

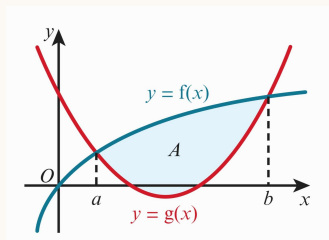


$$A = \int_a^b x \, dy \quad \cdot \text{area between curve and y-axis}$$



$$\Sigma A = R_1 + |R_2|$$

* area, bounded by $\begin{cases} \text{a curve + a line} \\ \text{2 curves} \end{cases}$



$$A = \int_a^b f(x) \, dx - \int_a^b g(x) \, dx$$

OR

$$A = \int_a^b [f(x) - g(x)] \, dx$$

7. Improper integrals : some part of the definite integral becomes infinite!

A) Type 1 : one or both limits are infinite

e.g. : $\int_1^{\infty} \frac{1}{x^2} \, dx$ replace ∞ with finite value X ,
then take the limit as $X \rightarrow \infty$

$$= \int_1^X x^{-2} \, dx$$

$$= [-x^{-1}]_1^X$$

$$= \left(-\frac{1}{X}\right) - \left(-\frac{1}{1}\right)$$

$$= 1 - \frac{1}{X}$$

As $X \rightarrow \infty$, $\frac{1}{X} \rightarrow 0$

$$\therefore \int_1^{\infty} \frac{1}{x^2} \, dx = 1 - 0 = 1$$

same process goes for $-\infty$ limit

B) Type 2 : the function itself approaches ∞ or $-\infty$ at one or both limits

e.g.: $\int_0^2 \frac{5}{x^2} dx \rightarrow$ at $x=0$, the function is undefined
as $x \rightarrow 0$, $\frac{1}{x^2} \rightarrow \infty$

$$\int_x^2 5x^{-2} dx$$

$$= [-5x^{-1}]_x^2$$

$$= \left(-\frac{5}{2}\right) - \left(-\frac{5}{x}\right)$$

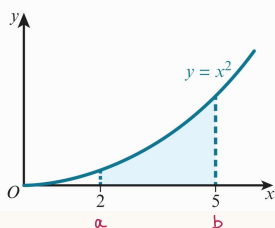
$$= \frac{5}{x} - \frac{5}{2}$$

replace with X ,
then take the limit as $X \rightarrow$ the replaced number

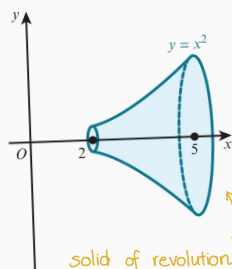
As $X \rightarrow 0$, $\frac{5}{X} \rightarrow$ infinity

$\Rightarrow$ function $\int_0^2 \frac{5}{x^2} dx$ is undefined

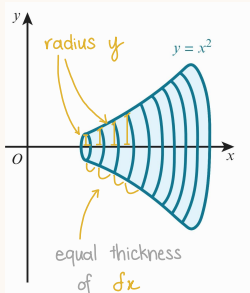
8. Volumes of revolution



rotated about x -axis
by 360°



divide volume into slices of equal thickness:



Approximation of V is $\sum \pi x y^2 \times \delta x$

As $\delta x \rightarrow 0$, then $V \rightarrow \int_2^5 \pi y^2 dx$

the total volume is divided
into equally thinner slices,
so thin! thickness is practically 0

for functions rotated
about x -axis:

$$V = \int_a^b \pi y^2 dx$$

for functions rotated
about y -axis:

$$V = \int_a^b \pi x^2 dy$$